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Task Description:

The research project was aimed at studying thermoacoustic and buoyancy-driven transport in gases and in fluids near their critical points at normal and reduced gravity environments. It is envisioned that supercritical fluid reactors will play a vital role in the development of 'water recovery' and waste treatment systems in long duration manned space missions. The present four-year project (start date: December 8, 2003) combined computational and experimental studies of thermoacoustic waves and thermoacoustic convection in gases and in near-critical fluids. In the duration of the project, significant advances were made in both the model development and the experimental design and facility development. Large-scale detailed computational models were developed and tested to simulate, and understand the thermoacoustically driven flows and transport in gases and the generation and propagation of thermoacoustic waves in supercritical carbon dioxide. Both normal-gravity and reduced-gravity experiments are planned to validate the key features of the simulations. For normal-gravity experiments in gases and in supercritical fluids, two experimental facilities have been designed and built at Drexel University. We extended our computational fluid dynamic models (for predicting thermoacoustic waves in supercritical fluids) by incorporating the NIST Reference Database 12 for supercritical carbon dioxide properties. In the case of low-heat-diffusing, near-critical fluids, it has been recently shown that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. The piston-effect has been successfully captured in our simulations using the models developed.
Normal gravity experiments were conducted for the generation and characterization of thermoacoustic waves in gases. Preliminary data agree well with model predictions. An experimental setup to generate and characterize the thermoacoustic waves in supercritical carbon dioxide has been designed and fabricated. Studies will be carried out under both normal and reduced gravity environments.

Regarding reduced-gravity experiments at the NASA Glenn 2.2 s drop tower, we visited the facilities in August 2005. Modified experimental setups were also designed that will satisfy the drop tower capabilities and guidelines. No drop tower experiments could be carried out due to the non-availity of the facility to the PI and his students. The available funding did not include the additional charges that would have incurred for the use of the facility as it was being closed.

The present research is aimed at developing advanced computational tools to analyze interactions between thermoacoustic waves and viscous fluids in several problems of significant relevance to microgravity conditions.

The specific objectives of the proposed research are the following:

• Improve and refine a high-order computational model for the simulation of flow and heat transfer in gases and near-critical fluids induced by thermoacoustic waves

• Perform direct numerical simulation of thermoacoustically driven gas flows in 2-D and 3-D enclosures

• Completely characterize the behavior of the thermoacoustic waves (generation, propagation, interaction with solid walls and eventual dissipation) in gases and nar-critical fluids

• Perform normal-gravity experiments for thermoacoustic waves in gases for model validation

• Perform reduced-gravity experiments (short-time) in drop-towers or KC 135 aircraft for isolating buoyancy effects from thermoacoustic wave generated flows in gases

• Select appropriate equation of state for near-critical fluids and incorporate the equation in the numerical model for flow simulation in such fluids

• Perform direct numerical simulation of the piston effect in near-critical fluids in enclosures

• Execute model for predicting the 'thermoacoustic' effects in near-critical fluids with and without gravitational effects

• Perform normal-gravity experiments with near-critical CO2 in carefully designed experiments (which reduce gravitational effects)

• Develop guidelines for the use of the mathematical models as a design and optimization tool for property and transport characterization in supercritical fluids and supercritical reactors.

Successful completion of the research tasks will provide us important information regarding the design and operation of supercritical reactors. Such reactors are envisioned for waste management and water recovery in long duration manned space missions.

The effects of thermoacoustic waves on transport and mixing in microgravity environments in such reactors willbe significant. The present research will be useful for the design of supercritical carbon dioxide reactors used in the food and pharmaceutical industries and for supercritical water reactors for waste management.

The presence of gravitational field leads to both practical and fundamental limits of the resolution in critical phenomena experiments in fluids near the gas-liquid critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. Near the critical point, fluids exhibit anomalies in the transport of heat such that convection and buoyancy phenomena appear for even minute temperature gradients. In the case of low-heat-diffusing, near-critical fluids, it has been shown recently that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. A significant question regarding the 'piston effect' is whether the fast heat transport is due to the strong buoyant convection induced by normal gravity in near-critical fluids or if it is due to the high compressibility of such fluids. Microgravity environment is thus ideal for studying such processes for fluids near the critical point where new fluid physics can be discovered.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Supercritical fluids exhibit a number of specific properties like large density, low viscosity, low thermal diffusivity. In addition, their isothermal compressibility and thermal expansion can become very large, especially when they approach the critical point. The highly variable properties of near-critical fluids make them very attractive for studying many interesting phenomena that hold for all fluids because of the critical universality. Fluids in their supercritical state are increasingly used by the food and waste management industry for their solubilization properties (e.g. supercritical carbon dioxide), as host of “cold” combustion (e.g. supercritical water), in energetics (supercritical thermal or nuclear power plants) and in astronautics (e.g. storage of cryogenic fluids).
It is particularly challenging to understand the behavior, flow and transport in a fluid that is close to its critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. The research results will provide valuable insights to the transport processes in supercritical reactors - whether they are used on earth or long duration manned space missions.

Task Progress:

We made significant progress in model development (for characterizing thermally generated acoustic waves in gases and near-critical fluids) and in the design and fabrication of two experimental set ups to study thermally generated acoustic waves in gases and in supercritical CO2 and N2. Experimental results were obtained and validated for thermally induced acoustic waves in gases in earth gravity environment. We plan to have additional experimental results on the characteristics of thermally induced acoustic waves in supercritical carbon dioxide and nitrogen by the end of summer 2009. This work was delayed due to the non-availability of a sensitive microphone that can withstand high pressure.
Computational studies:

(a) Studies in ideal gas

The effect of gravity on the generation and propagation of thermally generated acoustic wave induced flows in an enclosure was numerically investigated. The compressible unsteady Navier-Stokes equations were solved by a flux-corrected transport algorithm for the convection terms and by a central differencing scheme for the viscous and conduction terms. For the present simulations, the left wall of the enclosure is heated rapidly either in a spatially uniform or a non-uniform manner, while the right wall is held at the initial temperature of the gas. The horizontal walls of the enclosure are insulated. Under zero-gravity conditions, the spatially non-uniform heating is found to generate a vortical flow in the enclosure, similar to that found in buoyancy induced flows in a side-heated enclosure. For the conditions considered, the strength of the thermally generated acoustic wave induced flows is comparable to the buoyancy induced flows. Thermally generated acoustic wave induced flow fields were simulated using a 2D time-dependent compressible Navier-Stokes code.

(b) Studies in supercritical fluids

Under microgravity, heat transfer in supercritical fluids is expected to be very weak, due to the vanishing thermal diffusivity near the critical point. However, fast equilibration time was detected in past experiments because the bulk fluid is homogeneously heated by the thermally generated acoustic waves generated. This effect is known as the piston effect. The generation and propagation of a thermally generated acoustic wave in supercritical carbon dioxide was investigated by solving the fully compressible form of the Navier-Stokes equations. Two infinite plates filled with supercritical carbon dioxide, where the right wall is thermally insulated, is considered as the computational domain. A thermally induced pressure wave is generated by heating the left wall. The wave repeatedly traverses the length between the left and right walls, and its amplitude eventually damps out due to the viscous and thermal losses within the fluid, thus heating the bulk fluid. Long-time results show thermal equilibration and wave damping in the enclosure, which validates the existence of the piston effect in supercritical carbon dioxide.

Thermally generated acoustic wave generation and propagation in supercritical carbon dioxide was simulated using a one and two-dimensional compressible Navier-Stokes solver along with the NIST Reference Database 12 for the thermodynamic property relations. Computational models were developed for predicting the propagation and decay of thermoacoustic waves in supercritical nitrogen and carbon dioxide for 1- and 2-D geometries. The effect of gravity on the thermally generated acoustic waves and subsequent natural convection in supercritical fluids were predicted.

Experimental Studies: (a) Studies in ideal gas

An experimental facility was designed and built to study acoustically induced pressure waves in an enclosure. In the experimental setup, we could accurately measure the pressure waves (acoustic waves) due to sudden heating along an end wall. The measured data shows the generation and propagation of thermally generated acoustic waves in the enclosure. The wave reverberates within the enclosure and each peak indicates the passage of the wave after getting reflected from a side wall.

For the experimental system, (1) all separated parts were connected by flanges and rubber gaskets; (2) the chamber was thermally insulated; (3) gas filling and venting connectors were added; (4) absolute pressure within the chamber was also measured. With the designed experimental setup, we achieved the following goals: (a) eliminate the mass leakage and heat loss of the chamber; (b) run the experiments for different kinds of gas at different initial pressures.

Proposed microgravity experiments:

Following our trip to the NASA 2.2 s drop tower facilities at Cleveland, Ohio (August 7 – 9, 2005), and meetings with Mr. Eric Bauman, Manager 2.2 s drop tower facilities, and Ms. Nancy Hall, Research Scientist (and our program monitor), a new experimental setup for studying thermally generated acoustic waves in gases was designed. After the technical tour to the 2.2 s drop tower, we submitted a ‘support request’ to Mr. Bauman. According to the requirement of the Drop Tower, initial mechanical and electrical designs were made. To better understand the gravity effect on the thermally generated acoustic wave induced flow fields, we planned to add flow visualization experiments. We redesigning and improving our experimental setup. We also requested data acquisition hardware (a Generic Power Distribution Module, GPDM and a Digital Design and Custom Solution, DDACS data logger) commonly used for data acquisition in drop tower tests from NASA. The measurements with the GPDM and DDACS and flow visualization experiments were tested in the Hess Lab (Drexel University). Unfortunately, micro-gravity experiments were not carried out in the 2.2 s NASA drop tower for reasons mentioned earlier.

Due to the closing of the drop-tower facility during the later years of the project, the proposed experiments (despite our initial efforts and planning) could not be conducted. Since no additional amount was budgeted for the additional cost of

using the drop-tower facility (due to its closing for regular use), we abandoned our plans for the microgravity experiments.

(b) Studies in supercritical carbon dioxide

We used carbon dioxide for most of the experiments. CO2 is pumped from a tank (800 psi) into areactor to exceed the critical pressure of 7.38 MPa (1100 psi) value. A heating tape was used to increase the carbon dioxide temperature above the critical value (31.8ºC). The reactor is made of stainless steel. In one end of the reactor, we used a thin metal foil, which can be rapidly heated by a resistance-capacitance circuit, to produce thermally generated acoustic waves. We measure the instantaneous temperature by a fine-wire thermocouple and the pressure by a Bruel and Kjaer microphone. For safety, we employed several rupture discs in the charged lines and in the test chamber. The rupture pressure is set to be 10.34 MPa (1500 psi) for all the rupture discs, which allows us to do the experiments in pressure range of 7.38 – 10.34 MPa (1100 – 1500 psi). We also designed and built a polycarbonate cage as shown to house the test chamber. For the electrodes, thermocouples and the cable for the microphone, we use seals (connectors with glands) to prevent leakage. We used a machine thread for the end caps of the reactor, and we used Teflon gaskets to seal the end cap.

A special housing was constructed for mounting the microphone to the test chamber. As we were choosing the pressure sensor, we met a few challenges: microphones are rated for atmospheric pressure, and have only been tested up to 1.1 MPa (160 psi) not 7.38 MPa (1100 psi), the critical pressure of CO2. Hydrophones, which measure acoustic waves in high pressure fluid environments (such as sonar), were considered, however, they were found to be too bulky for the present setup. We needed to pressurize all components or isolate the pre-amplifier of the microphone from the high pressure zone, if we used a microphone to measure the pressure waves in supercritical carbon dioxide. We used an adaptor to isolate the preamplifier from the high pressure in the chamber.

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